

EIF-Based Inference for `misskappa` (`kappa_raw` backend)

This note records the influence-function (EIF) calculations used for asymptotic inference in the `kappa_raw()` “nonparametric” backend (the C++ `kappanp` code path), which produces Conger’s kappa, Fleiss’ kappa, and Brennan–Prediger (BP).

1. Data, missingness, and loss

For each subject $i \in \{1, \dots, n\}$ and rater $j \in \{1, \dots, R\}$, let X_{ij} denote a categorical rating taking values in $\{c_1, \dots, c_C\}$, and let $M_{ij} = 1[X_{ij} \text{ is observed}]$ indicate non-missingness. Write $X_i = (X_{i1}, \dots, X_{iR})$ and $M_i = (M_{i1}, \dots, M_{iR})$. We treat $\{(X_i, M_i)\}_{i=1}^n$ as i.i.d. across subjects.

Let $\ell(\cdot, \cdot)$ be a nonnegative *loss* (disagreement) function on categories. In the implementation, a weight matrix W is built first and the loss matrix is $L = \mathbf{1}\mathbf{1}^\top - W$, so $\ell(a, b) = L(a, b)$ and typically $\ell(a, a) = 0$.

2. Target functionals

The estimators are written in terms of *disagreement* components. Define two scalar ratios:

$$d = \frac{\psi_{dN}}{\psi_{dD}}, \quad d_C = \frac{\psi_{CN}}{\psi_{CD}}, \quad d_F = \frac{\psi_{FN}}{\psi_{FD}},$$

where the ψ ’s are expectations of subject-level or pairwise kernels defined below.

The three reported kappas are then smooth functions of (d, d_C, d_F) :

$$\kappa_C = 1 - \frac{d}{d_C}, \quad \kappa_F = 1 - \frac{d}{d_F}, \quad \kappa_{BP} = 1 - \frac{d}{d_{BP}},$$

where d_{BP} is treated as a known constant given the loss matrix:

$$d_{BP} = \frac{1}{C^2} \sum_{m=1}^C \sum_{m'=1}^C \ell(c_m, c_{m'}).$$

For the unweighted 0/1 loss $\ell(a, b) = 1[a \neq b]$, $d_{BP} = 1 - \frac{1}{C}$.

3. Sample estimators (raw ratings)

Observed disagreement: within-subject (U-statistic numerator/denominator)

Let $w_{jk}^{(\text{obs})}$ be a nonnegative weight assigned to the within-subject pair (j, k) (distinct raters, $j < k$). Define the subject-level contributions

$$h_{dN}(X_i, M_i) = \sum_{1 \leq j < k \leq R} M_{ij} M_{ik} \ell(X_{ij}, X_{ik}) w_{jk}^{(\text{obs})},$$

$$h_{dD}(X_i, M_i) = \sum_{1 \leq j < k \leq R} M_{ij} M_{ik} w_{jk}^{(\text{obs})}.$$

Then $\psi_{dN} = \mathbb{E}[h_{dN}]$ and $\psi_{dD} = \mathbb{E}[h_{dD}]$ and the sample analogs are

$$\hat{\psi}_{dN} = \frac{1}{n} \sum_{i=1}^n h_{dN}(X_i, M_i), \quad \hat{\psi}_{dD} = \frac{1}{n} \sum_{i=1}^n h_{dD}(X_i, M_i), \quad \hat{d} = \frac{\hat{\psi}_{dN}}{\hat{\psi}_{dD}}.$$

Chance disagreement: cross-subject (V-statistic numerators/denominators)

Let $w_{jk}^{(\text{exp})}$ be a nonnegative weight for cross-subject pairings between rater j on subject i and rater k on subject i' . Define four kernels:

$$K_{CN}((X_i, M_i), (X_{i'}, M_{i'})) = \sum_{1 \leq j < k \leq R} M_{ij} M_{i'k} \ell(X_{ij}, X_{i'k}) w_{jk}^{(\text{exp})},$$

$$K_{CD}((X_i, M_i), (X_{i'}, M_{i'})) = \sum_{1 \leq j < k \leq R} M_{ij} M_{i'k} w_{jk}^{(\text{exp})},$$

$$K_{FN}((X_i, M_i), (X_{i'}, M_{i'})) = \sum_{j=1}^R \sum_{k=1}^R M_{ij} M_{i'k} \ell(X_{ij}, X_{i'k}) w_{jk}^{(\text{exp})},$$

$$K_{FD}((X_i, M_i), (X_{i'}, M_{i'})) = \sum_{j=1}^R \sum_{k=1}^R M_{ij} M_{i'k} w_{jk}^{(\text{exp})}.$$

The **misskappa** implementation uses the corresponding V-statistic estimators (including $i = i'$ terms):

$$\hat{\psi}_{CN} = \frac{1}{n^2} \sum_{i=1}^n \sum_{i'=1}^n K_{CN}((X_i, M_i), (X_{i'}, M_{i'})), \quad \hat{\psi}_{CD} = \frac{1}{n^2} \sum_{i=1}^n \sum_{i'=1}^n K_{CD}((X_i, M_i), (X_{i'}, M_{i'})),$$

and similarly $(\hat{\psi}_{FN}, \hat{\psi}_{FD})$. The chance disagreements are $\hat{d}_C = \hat{\psi}_{CN}/\hat{\psi}_{CD}$ and $\hat{d}_F = \hat{\psi}_{FN}/\hat{\psi}_{FD}$.

4. EIF / influence functions

Define the 6-vector of ψ components

$$\psi = (\psi_{dN}, \psi_{dD}, \psi_{CN}, \psi_{CD}, \psi_{FN}, \psi_{FD})^\top, \quad \hat{\psi} = (\hat{\psi}_{dN}, \hat{\psi}_{dD}, \hat{\psi}_{CN}, \hat{\psi}_{CD}, \hat{\psi}_{FN}, \hat{\psi}_{FD})^\top.$$

U-statistic components

Because $\hat{\psi}_{dN}$ and $\hat{\psi}_{dD}$ are simple averages of subject-level terms, their influence functions (for the i.i.d. subject model) are

$$\phi_{dN}(X_i, M_i) = h_{dN}(X_i, M_i) - \psi_{dN}, \quad \phi_{dD}(X_i, M_i) = h_{dD}(X_i, M_i) - \psi_{dD}.$$

In practice we replace ψ_{dN}, ψ_{dD} by their sample estimates.

V-statistic components

For a generic V-statistic of the form

$$\hat{\psi}_K = \frac{1}{n^2} \sum_{i=1}^n \sum_{i'=1}^n K(Z_i, Z_{i'}), \quad Z_i = (X_i, M_i),$$

the first-order (Hoeffding) influence-function term can be written as

$$\phi_K(Z_i) = \mathbb{E}[K(Z_i, Z')] + \mathbb{E}[K(Z', Z_i)] - 2\psi_K.$$

The implementation uses the empirical versions

$$\hat{g}_{K,1}(Z_i) = \frac{1}{n} \sum_{i'=1}^n K(Z_i, Z_{i'}), \quad \hat{g}_{K,2}(Z_i) = \frac{1}{n} \sum_{i'=1}^n K(Z_{i'}, Z_i),$$

$$\hat{\phi}_K(Z_i) = (\hat{g}_{K,1}(Z_i) - \hat{\psi}_K) + (\hat{g}_{K,2}(Z_i) - \hat{\psi}_K).$$

Apply this with $K \in \{K_{CN}, K_{CD}, K_{FN}, K_{FD}\}$ to obtain $(\hat{\phi}_{CN}, \hat{\phi}_{CD}, \hat{\phi}_{FN}, \hat{\phi}_{FD})$.

Stacked influence vector and covariance

Let

$$\hat{\phi}(Z_i) = (\hat{\phi}_{dN}(Z_i), \hat{\phi}_{dD}(Z_i), \hat{\phi}_{CN}(Z_i), \hat{\phi}_{CD}(Z_i), \hat{\phi}_{FN}(Z_i), \hat{\phi}_{FD}(Z_i))^\top.$$

The covariance of $\hat{\psi}$ is estimated by

$$\hat{\Gamma} = \frac{1}{n} \sum_{i=1}^n \hat{\phi}(Z_i) \hat{\phi}(Z_i)^\top, \quad \widehat{\text{Var}}(\hat{\psi}) \approx \frac{1}{n} \hat{\Gamma}.$$

5. Delta method to (d, d_C, d_F) and to kappas

From ψ to disagreements

Let $d(\psi) = (\psi_{dN}/\psi_{dD}, \psi_{CN}/\psi_{CD}, \psi_{FN}/\psi_{FD})^\top$. Its Jacobian is the 3×6 matrix

$$J_d(\psi) = \begin{pmatrix} \psi_{dD}^{-1} & -\psi_{dN}\psi_{dD}^{-2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \psi_{CD}^{-1} & -\psi_{CN}\psi_{CD}^{-2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \psi_{FD}^{-1} & -\psi_{FN}\psi_{FD}^{-2} \end{pmatrix}.$$

Plugging in $\hat{\psi}$ yields $\hat{J}_d = J_d(\hat{\psi})$ and

$$\widehat{\text{Var}}((\hat{d}, \hat{d}_C, \hat{d}_F)^\top) \approx \frac{1}{n} \hat{J}_d \hat{\Gamma} \hat{J}_d^\top.$$

From disagreements to kappas

Let $\kappa(d) = (1 - d/d_C, 1 - d/d_F, 1 - d/d_{BP})^\top$, where d_{BP} is constant. The Jacobian of κ with respect to (d, d_C, d_F) is

$$J_\kappa(d) = \begin{pmatrix} -d_C^{-1} & d d_C^{-2} & 0 \\ -d_F^{-1} & 0 & d d_F^{-2} \\ -d_{BP}^{-1} & 0 & 0 \end{pmatrix}.$$

Plugging in $(\hat{d}, \hat{d}_C, \hat{d}_F)$ gives $\hat{J}_\kappa$ and

$$\widehat{\text{Var}}(\hat{\kappa}) \approx \hat{J}_\kappa \widehat{\text{Var}}((\hat{d}, \hat{d}_C, \hat{d}_F)^\top) \hat{J}_\kappa^\top.$$

6. Weight choices and relation to method

Let

$$\hat{\pi}_j = \frac{1}{n} \sum_{i=1}^n M_{ij}, \quad \hat{\pi}_{jk} = \frac{1}{n} \sum_{i=1}^n M_{ij} M_{ik}.$$

The code path underlying `kappa_raw(method = ...)` differs only by the choices of $w_{jk}^{(\text{obs})}$ and $w_{jk}^{(\text{exp})}$:

$$w_{jk}^{(\text{obs})} = \begin{cases} 1 & \text{"available"} \\ \hat{\pi}_{jk}^{-1} & \text{"ipw"} \\ 1 & \text{"gwet"} \end{cases} \quad w_{jk}^{(\text{exp})} = \begin{cases} 1 & \text{"available"} \\ \hat{\pi}_j^{-1} \hat{\pi}_k^{-1} & \text{"ipw"} \\ \hat{\pi}_j^{-1} \hat{\pi}_k^{-1} & \text{"gwet"} \end{cases}.$$

This matches `calculate_inverse_weights()` in `code/misskappa/src/kappanp.cpp`.

7. Remarks for categorical data and implementation

- **Raw format and Cohen’s kappa.** The “raw” data format stores per-rater ratings, which enables pairwise comparisons. When $R = 2$, Conger’s kappa specializes to a Cohen-style kappa computed from the single rater pair $(1, 2)$; Fleiss’ chance term averages over all ordered rater pairs and can differ from Cohen unless raters have (approximately) the same marginal rating distribution.
- **Finite support (pattern compression).** For categorical ratings with missingness, Z_i takes values in a finite set of rating/missingness patterns. The implementation exploits this by computing h_{dN}, h_{dD} and the K kernels once per unique pattern and expanding back to per-subject influence values; this is computational and does not change the formulas above.
- **Plug-in standard errors.** Once $\widehat{\text{Var}}(\hat{\kappa})$ is available, standard errors are $\widehat{se}(\hat{\kappa}_r) = \sqrt{\widehat{\text{Var}}(\hat{\kappa})_{rr}}$ and Wald intervals follow in the usual way.
- **Degeneracy checks.** The denominators $\hat{\psi}_{dD}, \hat{\psi}_{CD}, \hat{\psi}_{FD}$ (and d_{BP}) must be non-negligible; the code guards against near-zero denominators by returning NaN kappas when needed.